Reader: What else do we need?

Author: Do you think that any number could be fed into a specific "black box" (function)?

Reader: I see. I have to define a set of numbers acceptable as inputs of the given function.

Author: That's right. This set is said to be the **domain of a function**. Thus, the definition of a numerical function is based on two "cornerstones": the domain of a function (a certain set of numbers), and the law of numerical correspondence.

According to this law, every number from the domain of a function is placed in correspondence with a certain number, which is called the value of the function; the values form the range of the function. V. Imp

Reader: Thus, we actually have to deal with two numerical sets. On the one hand, we have a set called the domain of a function and, on the other, we have a set called the range of a function.

Author: At this juncture we have come closest to a mathematical definition of a function which will enable us to avoid the somewhat mysterious word "black box".

- Look at Fig. 13. It shows the function $y=\sqrt{1-x^2}$. Figure 13 pictures two numerical sets, namely, D (represented by the interval $[-1,1]$) and E (the interval $[0,1]$). For your convenience these sets are shown on two different real lines.

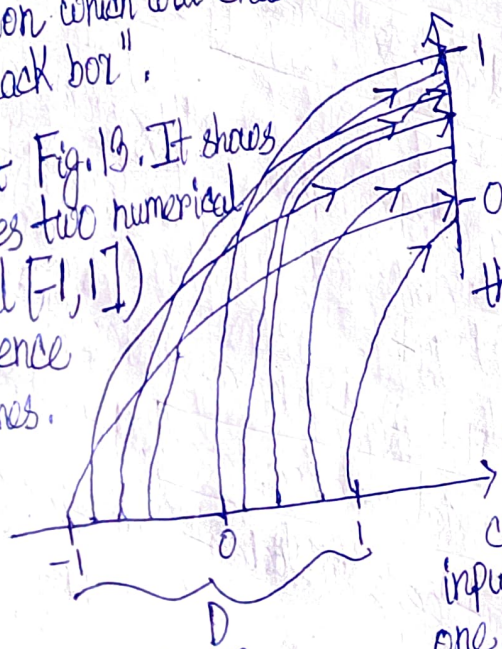

Fig. 13

- The set D is the domain of the function, and E is its range. Each number in D corresponds to one number in E (every input value is placed in correspondence with one output value). This correspondence is shown in Fig. 13 by arrows pointing from D to E.